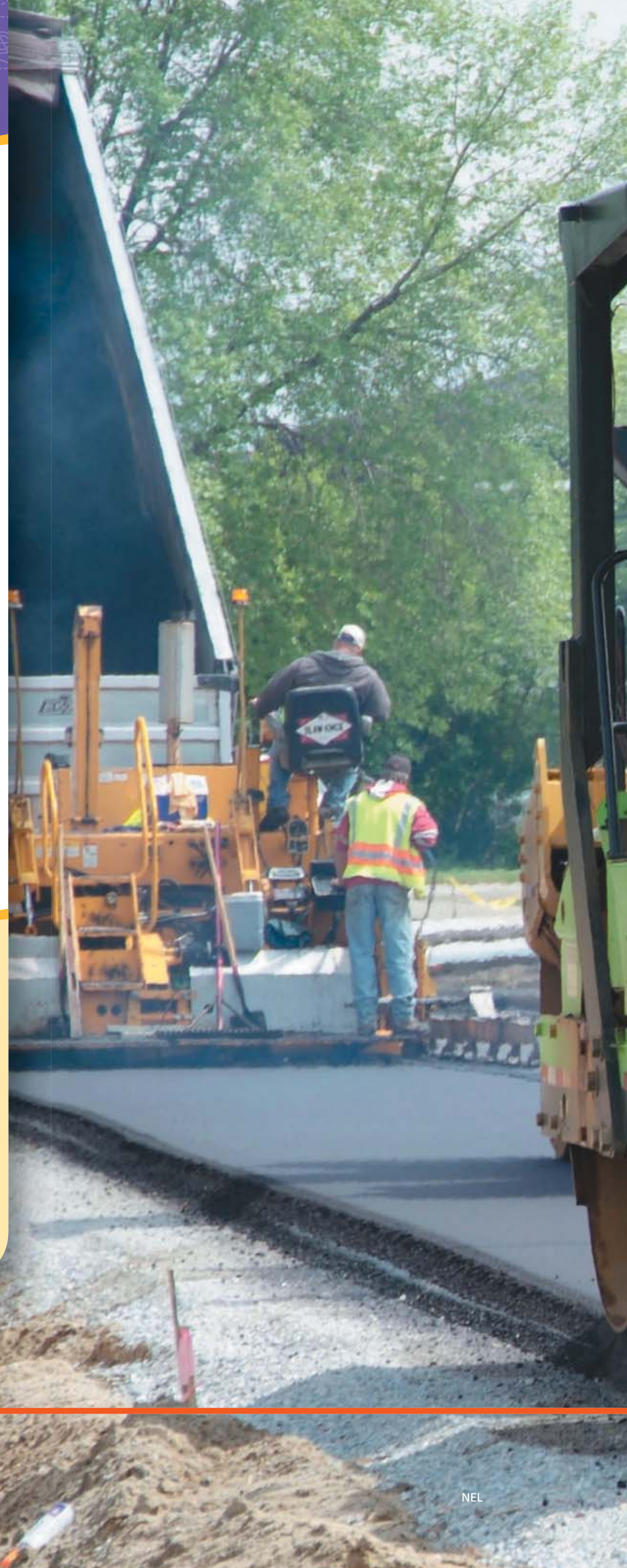
UNIT 2

Chemical processes require energy change as atoms are rearranged.

Many chemical compounds in the world around you are made by living things as part of their life processes. The sugars made by green plants are one example. People also isolate and make (synthesize) chemical compounds in laboratories and in industrial settings. Some synthetic compounds, such as the asphalt used to pave roads, are found in nature. Others, such as the plastics used to make traffic cones and hard hats, come from natural sources, but they do not exist in nature.

“Green chemistry is replacing our industrial chemistry with nature’s recipe book. It’s not easy, because life uses only a subset of the elements in the periodic table. And we use all of them, even the toxic ones.”

Janine Benyus
Natural sciences author, innovation consultant



- What role does energy play in the production of chemical substances in nature and those we synthesize?
- What are the benefits and the costs of the chemical substances and products that we synthesize?
- What questions do you have about—this photo? the title of this unit? …?

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UNIT 2

At a Glance

You will demonstrate what you know, can do, and understand by being able to

- Perform investigations and use other investigative methods to explore properties and patterns involving elements and compounds
- Use scientific understandings to describe and explain the role of energy in chemical processes
- Seek patterns and connections to identify and describe types of chemical reactions
- Develop increasing facility to express ideas, attitudes, and actions in respectful and responsible ways

ESSENTIAL QUESTION

What happens to the energy and atoms of substances in chemical reactions?

TOPIC 2.1:

How are chemical processes part of our lives?

Some things you will do:

- demonstrate sustained intellectual curiosity about a topic or problem of personal interest
- ensure that safety and ethical guidelines are followed in investigations

Some things you will come to know:

- You live in a world filled with applications of chemistry.
- You have a responsibility to use chemicals and chemical knowledge safely.

TOPIC 2.2:
What happens to atoms in a chemical reaction?

Some things you will do:

- apply First Peoples perspectives and knowledge, other ways of knowing, and local knowledge as sources of information
- construct, analyze, and interpret models and diagrams
- transfer and apply learning to new situations

Some things you will come to know:

- The making and breaking of chemical bonds involves changes in energy.
- Writing chemical equations depends on the law of conservation of mass.

TOPIC 2.3:
How is energy involved in chemical processes?

Some things you will do:

- collaboratively and individually plan, select, and use appropriate investigation methods
- analyze cause-and-effect relationships
- contribute to care for self, others, community and world through individual and collaborative approaches

Some things you will come to know:

- Chemical reactions involve a transfer of energy between systems and their surroundings.
- Some chemical reactions absorb energy, and others release energy.

TOPIC 2.4:
How do atoms rearrange in different types of chemical reactions?

Some things you will do:

- seek and analyze patterns, trends, and connections in data
- communicate scientific ideas, claims, information, and courses of action for a specific purpose and audience

Some things you will come to know:

- The many kinds of chemical reactions can be grouped into a few main types based on how their atoms are rearranged.
- Acids and bases are common chemical substances in daily life.

UNIT 2

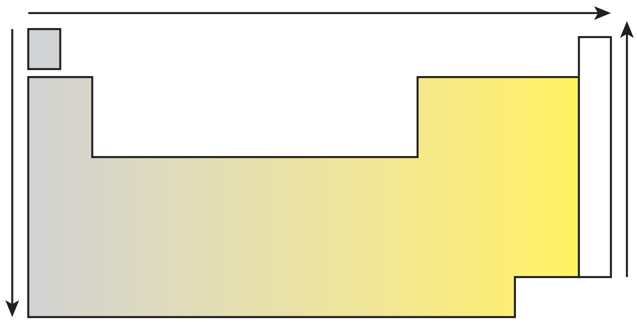
Connect To What You Already Know

This feature helps you reflect on what you know about some foundational ideas that you have learned in previous grades. Work alone or quietly in small groups to answer the questions. Reach out to your classmates to ask about things that you are unsure about or to offer assistance. Your teacher also can provide additional reinforcement materials to help you prepare for this unit.

Refer to **Figure 1** on the next page as well as the periodic table on the inside back cover of this textbook as you answer the following questions.

1. Look at the symbols and atomic numbers of the elements.
   a) What is the atomic number of helium?
   b) What is the atomic number of gold?
   c) What is the symbol of the element with atomic number 22?
   d) What is the symbol of the element with atomic number 33?
2. Look at the atomic masses of the elements.
   a) What is the atomic mass of aluminum?
   b) What is the atomic mass of silver?
   c) What is the symbol of the element with atomic mass 40.1?
   d) What is the symbol of the element with atomic mass 83.8?
3. What is the difference between an element's atomic number and its atomic mass? What is the significance of the atomic number in terms of the periodic table?

Figure 1



4. Copy the diagram below into your notebook, and use it to answer the following.
   a) What physical properties do the grey-shaded elements have in common?
   b) What physical properties do the yellow-shaded elements have in common?
   c) What physical properties do the elements in the white column have in common?
   d) One element appears on its own in the upper left corner of the periodic table. What makes this element unique in terms of many of its physical properties?
   e) Identify the trends in physical and chemical properties represented by the arrows that surround **Figure 1**.
5. Elements 1, 3, 11, and 19 are in the first column of the periodic table.
   a) What are the two terms used to identify columns of the periodic table?
   b) Draw Bohr diagrams for these elements.
   c) How is the electron arrangement in these elements similar?
   d) How many electrons would you expect there to be in the outer energy shell of the elements Rb and Cs? Explain.
6. Look at elements 3 to 10.
   a) Draw Bohr diagrams for these elements
   b) Compare your diagrams. How are they the same? How are they different?
   c) What period do these elements belong to?
   d) What is the relationship between period number and occupied energy shells?

7. An element's reactivity is associated with how close it is to having a full valence shell.
   a) What elements have full valence shells? Are they the most or least reactive elements? Explain.
   b) Why are elements of Group 1 and 17 the most reactive elements?
   c) What is an ion? Using Bohr diagrams of representative elements, show how a negatively charged and a positively charged ion forms. Explain the formation of ions for these elements.

8. Elements in the same column of the periodic table have similar properties. Think of two examples in everyday life where similar substances could be substituted for each other. What other factors would you consider before making the substitutions?

9. a) Which elements on the periodic table do not occur naturally on Earth?
   b) Of these, which ones are synthetic (made only in the laboratory)?
   c) In both cases, what is it about the nature of these elements that accounts for the fact that they are not naturally occurring on Earth?

Figure 2

Elements are classified as metals, non-metals, and semi-metals based on their physical and chemical properties. The elements in groups 1, 2, and 13 to 18 are referred to as the main-group elements or representative elements. The elements in groups 3 to 12 are called the transition elements.

Most of the elements are metals. With the exception of mercury, metals are solid at room temperature. They are shiny when smooth and clean, and most are silver or grey in colour. They are good conductors of electric current and thermal energy. They are also malleable and ductile.

The non-metals, except for hydrogen, are found on the upper right side of the periodic table. Non-metals are general gases or brittle, dull-looking solids. They are poor conductors of electric current and thermal energy.

Semi-metals (also called metalloids) have physical and chemical properties of both metals and non-metals. For example, like metals, they are shiny solids at room temperature, but they are brittle and not ductile like non-metals. They also tend to be poor conductors of electric current and thermal energy.